

TOPAZ: Establishing Topology Optimization as a Paradigm for Passive Noise-Resilient Quantum Computing

Krishanu Karun* ¹

¹Independent Researcher

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Abstract

This paper introduces the Continuous Topological Interpretation (CTI), which reframes quantum circuits as continuous landscapes embedded in an extended parameter space, and shows that entanglement acts as a mechanism of manifold enlargement while noise behaves as geometric deformation. Building on this foundation, we develop TOPAZ, a noise-resilient topology optimization framework that pre-tunes ansatz parameters before hardware execution. TOPAZ combines a Parametric Trotter Expression (PTE) with polar projection for expressivity, a dual-MMA optimiser that decouples intensity and timing subspaces, and an $O(2^N)$ tensor-dot noise engine. Validation spans three scales: 12-qubit LiH (beating Zero-Noise Extrapolation by $1.87\times$), 8-qubit calibrated-device benchmarks (density-matrix fidelity 0.9085 with 21 two-qubit gates), and live IBM Heron r2 hardware (inversion test gain of $10.71\times$ at 18σ). A direct-optimisation ablation confirms the full stack is necessary. Across every backend, the optimiser converges to a hardware-transferable fidelity floor—direct empirical evidence for the “structural well” predicted by CTI.

Keywords: Continuous Topological Interpretation, Entanglement Geometry, Parametric Trotter Expression, Quantum Topology Optimization, Noise-Resilient Quantum Circuits, NISQ, IBM Heron, Structural Well, Method of Moving Asymptotes.

1 Introduction

The discrete-model bottleneck. In classical computer science, the most fundamental abstraction is the instruction sequence: a list of discrete operations executed in a rigid, step-by-step order. Early quantum computing has inherited this discrete mental model — a circuit is a finite sequence of unitary gates drawn from a discrete set and applied over discrete time steps.

While this view is indispensable for compilers and error-correcting codes, it creates bottlenecks when we optimize quantum circuits on physical hardware. In Variational Quantum Algorithms (VQAs) and Quantum Optimal Control (QOC), we optimize continuous physical parameters (laser pulse durations, gate rotation angles). But the gate-based representation confines search to the curved, non-convex manifold of $U(2^N)$. The resulting landscapes suffer from sharp energy cliffs, local minima, and “barren plateaus” [14] where gradient signals vanish exponentially with system size.

This paper introduces a different conceptual and mathematical framework: the *Continuous Topological Interpretation (CTI)* of quantum circuits. Rather than treating a quantum circuit as a static, discrete sequence of rigid operations, we view it as a continuous, high-dimensional landscape. In this geometric framework,

we embed the standard unitary group $U(d)$ as a constrained, physically realizable submanifold sitting inside a larger, unconstrained “ambient” space of general complex matrices, $GL(d, \mathbb{C})$.

[Insert illustration: CTI paradigm shift — discrete gates vs. continuous landscape]

Figure 1. Comparison between traditional discrete gate-based models and the Continuous Topological Interpretation.

By allowing our optimization parameters to temporarily step off the rigid unitary path and explore this larger ambient space, the mathematical equations governing our gradients become remarkably smooth. Gradients can flow freely around localized energy barriers that would block a strictly unitary optimizer. Prior to

physical implementation, we restore physical realizability by projecting the ambient operators back to the nearest unitary matrix using a mathematically clean polar projection.

By reframing quantum circuits as continuous landscapes, we establish three core theoretical results:

1. **Entanglement as Manifold Enlargement:** We prove that quantum entanglement has a clean geometric interpretation. It acts as a topological mechanism that physically enlarges the dimension of the accessible parameter manifold.
2. **Unitary Regularization:** We demonstrate that projecting ambient operators back onto the unitary group smooths the optimization landscape, stabilizing gradient signals.
3. **Noise Deformation:** We show that environmental noise can be viewed as a continuous vector field pulling the state landscape away from our target manifold.

Quantum error correction is not available on near-term hardware. What is available is a stack of quantum error mitigation (QEM) techniques — Zero-Noise Extrapolation (ZNE) [8], Probabilistic Error Cancellation [9], Clifford Data Regression (CDR) [10], dynamical decoupling [21], and twirled readout [20] — all of which operate on circuit outputs after execution. They post-process measurements, fit error models, or extrapolate from amplified-noise runs [17].

TOPAZ v1 [2] introduced an additional layer that acts *before* the circuit runs. Given a parametric Trotter ansatz with intensity weights ρ and timing angles τ , the framework holds the circuit’s gate topology fixed and tunes (ρ, τ) so the noisy output is intrinsically closer to the ideal one. The v1 paper established the PTE parameterisation [3], the triple-channel correlated noise model, the specialised Method of Moving Asymptotes (MMA) adaptation [4], and 8-qubit validation under simulated noise. It also identified Grover’s algorithm as incompatible with the framework on topological grounds (continuity and convexity violations in the amplitude-amplification landscape).

This paper is the follow-up. We retain the earlier mathematical structure and add what was needed to take the framework from simulator demonstrations to a calibrated device-noise model and to live hardware:

- A *dual-MMA* optimiser (Sec. 4.3) with two meanings: subspace separation between ρ and τ inside one module, and stacked MMA-over-PTE modules across parameter classes. From random initialisation, MMA contributes $\sim 4\%$ DM fidelity beyond Adam, confirming that the full architecture-init-optimiser stack is necessary; from analytic initialisation both tie at $DM \approx 0.91$.
- A tensor-dot noise engine (Sec. 4.4) that brings Kraus-map evaluation from $O(2^{3N})$ to $O(2^N)$ per term, with explicit scope: where the engine runs and where it does not.

- A hardware execution path (Sec. 4.5). The SVD-of-sum unitary used in simulator runs does not transpose cleanly for $N \geq 6$; we replace it on hardware with a product-of-exponentials Trotter form that compiles to a shallow 21-gate circuit for the 8-qubit case.
- A 12-qubit LiH benchmark, a head-to-head against ZNE/CDR/VD on the same physical baseline (Sec. 5.4), a topological-lift result for Grover’s diffusion oracle (Sec. 5.6) and Bernstein–Vazirani (Sec. 5.6), and a real-hardware run on IBM’s 156-qubit Heron-r2 (Sec. 5.8).
- A direct empirical observation of the “structural well” predicted by the companion CTI paper [1]: the optimiser converges to the same fidelity floor from random initialisations, and the floor transfers across hardware (Sec. 6.2). The well is not a simple 3D attractor but a sparse subspace (8–24 effective dimensions); its survivors are not the target state’s largest coefficients but those selected for noise-robustness, and the sparsity mechanism was not anticipated by CTI-E.

We retain the original positioning: TOPAZ is one layer of the QEM stack, not a stand-alone solution. Its job is to deliver a parameter setting that survives the device’s noise as well as possible before any post-execution mitigation runs. ZNE, DD, TREX and PEC consume the resulting circuit and benefit from its cleaner baseline.

VQA landscape: expressivity versus depth.. Variational Quantum Algorithms (VQAs) [6] are the leading approach for near-term quantum computing, but they face a fundamental tension: shallow circuits limit expressivity, yet coherence times demand shallow circuits. Traditional VQA parameterisations — fixed gate structures with rotation angles — can only explore limited subspaces of the unitary group. Maximising expressivity per gate is therefore critical.

PTE: a different parameterisation.. This work introduces the Parametric Trotter Expression (PTE), which expands expressivity through weighted superpositions of evolution primitives. Unlike conventional approaches that maintain strict unitarity at every step, PTE allows the operator to temporarily leave the unitary group, explore a larger manifold, and then project back via the polar decomposition. The result is the Unitary PTE (U-PTE) operator. The contributions here are: (1) theoretical derivation from Schrödinger evolution, (2) proof that weighted unitary sums require polar projection, (3) numerically stable construction algorithms, and (4) quantum circuit objective functions.

Note.. This paper validates the mathematical rigor of PTE. Implementation studies and practical benchmarks are addressed in companion work.

2 Theory (CTI): A Continuous Topological Interpretation of Quantum Circuits and Entanglement

2.1 The Ambient Space and the Unitary Submanifold

To build the mathematical foundation of CTI, we must define the space in which our continuous quantum operators reside.

The Extended Operator Space. Let $\mathcal{H}_N \cong \mathbb{C}^d$ be the Hilbert space of an N -qubit system, where $d = 2^N$. The standard operations are represented by the unitary group:

$$U(d) = \{U \in \text{Mat}_d(\mathbb{C}) : U^\dagger U = I\} \quad (1)$$

This group is a compact, real Lie group of dimension d^2 . In CTI, we embed $U(d)$ as a constrained submanifold within $GL(d, \mathbb{C})$, or more generally, the full space of linear operators $\text{Mat}_d(\mathbb{C}) \cong \mathbb{C}^{d^2}$. We define a continuous circuit operator $A(\lambda)$ as a mapping:

$$A : \mathbb{R}^M \rightarrow \text{Mat}_d(\mathbb{C}) \quad (2)$$

where λ represents the continuous coordinates of our landscape. Because $A(\lambda)$ is not restricted to be unitary, operations can interpolate and scale.

Polar projection.. To execute these operators on hardware, we map them back to the physically realizable unitary submanifold. By the polar decomposition theorem, every invertible matrix $A \in GL(d, \mathbb{C})$ has a unique decomposition $A = QP$ where Q is unitary and $P = (A^\dagger A)^{1/2}$ is positive definite. Q is the unique closest unitary to A in the Frobenius norm: $\mathcal{P}_U(A) = A(A^\dagger A)^{-1/2}$.

Unitary regularization.. This projection is the geometric bridge of the CTI framework. It guarantees that our algorithm can navigate smoothly in the unconstrained ambient space while ensuring the output $U_{\text{safe}}(\lambda) = \mathcal{P}_U(A(\lambda))$ is unitary. This process, **unitary regularization**, “pins” singular values back to 1.0, acting as a filter that smooths sharp wrinkles in the loss landscape and stabilizes gradient signals through deep circuits.

2.2 Quantum Entanglement as Manifold Enlargement

The “paper sheet” metaphor.. Let us visualize this using an intuitive geometric analogy. If we restrict our qubits to be unentangled, the states we can prepare form a thin, folded, low-dimensional sheet winding through the massive Hilbert space volume. We call this the unentangled manifold, $\mathcal{M}_{\text{unt}}$. Navigating $\mathcal{M}_{\text{unt}}$ is restrictive. Adding entangling operations is equivalent to physically inflating this thin sheet into a high-dimensional volume: the entangled manifold, $\mathcal{M}_{\text{ent}}$. Suddenly, the optimizer has extra dimensions of freedom to bypass bottlenecks.

[Insert illustration: entanglement as manifold inflation — paper sheet $\rightarrow$ high-dimensional volume]

Figure 2. Visualizing entanglement as a dimensional inflation of the landscape.

Proof of parameter-space enlargement.. Suppose we have an N -qubit register partitioned into k disjoint clusters S_a . Without entanglement, operations are restricted to localized rotations within each cluster. For each cluster S_a , we define:

$$U_{S_{\text{local},a}}(\lambda_{\text{local},a}) = \bigotimes_{j \in S_a} \exp\left(-i \frac{\theta_{a,j}}{2} \sigma_{j,a}\right) \quad (3)$$

The set of all preparable product states forms $\mathcal{M}_{\text{unt}}$:

$$\mathcal{M}_{\text{unt}} = \left\{ \bigotimes_{a=1}^k |\phi_a(\lambda_{\text{local},a})\rangle : \lambda_{\text{local},a} \in \mathbb{R}^{d_{\text{local},a}} \right\} \quad (4)$$

The dimension is bounded by $\dim(\mathcal{M}_{\text{unt}}) = \sum_{a=1}^k d_{\text{local},a}$. Now, introduce entangling parameters $\lambda_{\text{ent},a}$ that scale interaction terms bridging cluster boundaries. The combined circuit block dimension becomes $d_a = d_{\text{local},a} + d_{\text{ent},a}$. The dimension of the accessible landscape $\mathcal{M}_{\text{ent}}$ is now:

$$\dim(\mathcal{M}_{\text{ent}}) = \sum_{a=1}^k (d_{\text{local},a} + d_{\text{ent},a}) > \dim(\mathcal{M}_{\text{unt}}) \quad (5)$$

This confirms that **quantum entanglement physically inflates the dimension of our parameter space**, unlocking higher-dimensional navigation.

2.3 Optimizing over the Ambient Landscape

In traditional algorithms, optimization is performed directly on $U(d)$. This requires calculating gradients along curved directions. In CTI, we open up the ambient space $GL(d, \mathbb{C})$, where operations are linear and gradients are smooth. This unconstrained exploration allows CTI to navigate complex landscapes with speed. The polar projection acts as a guide rail, ensuring valid physical states while the ambient space provides the freedom to find the shortest pathway.

2.4 Noise as Geometric Deformation

We visualize noise as a physical deformation of our state landscape. Environmental noise (T_1, T_2) acts as an unwanted **continuous vector field** pulling the state toward high-entropy interiors. Instead of adding qubits,

Ambient optimization with polar projection

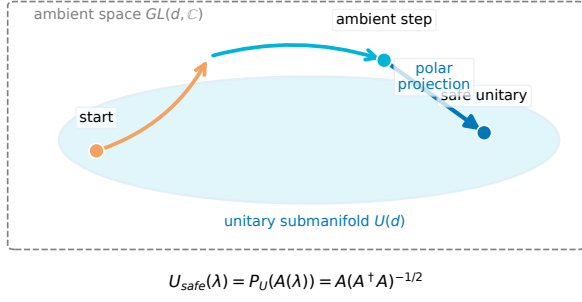


Figure 3. The optimization pathway in CTI leverages non-unitary space.

Noise as geometric deformation

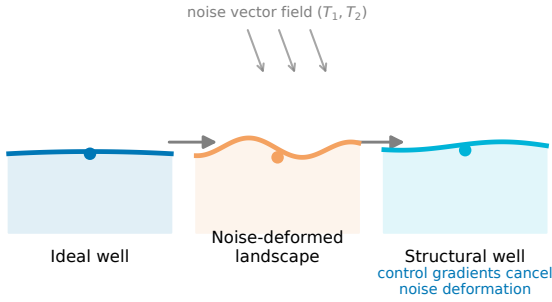


Figure 4. Visualizing noise as deformation and mitigation as landscape shaping.

we use coordinates λ to search for a **structural well** where the noise field and control gradients cancel. By pre-tuning parameters to sit inside this well, we build resilience directly into the circuit’s structural parameters.

3 The Operator (PTE/UPTE): Derivation and Validation of the Parametric Trotter Expression

3.1 Background

Motivation.. The Parametric Trotter Expression (PTE) is a parameterized operator ansatz where the free variables correspond to gate intensities and timings. The key purpose of PTE is not to automatically enforce unitarity, but to provide a flexible ansatz that allows an optimizer to explore the operator-level control landscape with unprecedented expressivity.

Theoretical Insight: PTE uses weighted superpositions of candidate sub-circuits, significantly expanding the reachable manifold beyond traditional variational quantum circuits. However, this expressivity comes at a cost: The resulting operators may stray outside the unitary manifold. The projection to the nearest unitary (the U-PTE operator) acts as a mathematical safeguard, not as the original motivation for introducing PTE.

The theoretical foundation rests on the observation that quantum hardware trends toward larger qubit

counts but shallower coherence windows. PTE addresses circuit depth by allowing weighted, timing-aware superpositions that approximate target unitaries with fewer gate layers.

From Schrödinger equation to time evolution.. The time-dependent Schrödinger equation governs quantum dynamics:

$$-i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = H |\Psi(t)\rangle \quad (6)$$

Formal Solution: Integrating this differential equation yields:

$$|\Psi(t)\rangle = U(t) |\Psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar} \quad (7)$$

Setting $\hbar = 1$ for convenience:

$$U(t) = e^{-iHt} \quad (8)$$

Theoretical Significance: The exponential operator e^{-iHt} encodes continuous time evolution. Simulating quantum dynamics means approximating this operator, establishing the mathematical bridge between continuous evolution and discrete approximation schemes.

Hamiltonian decomposition and non-commutativity.. We decompose the Hamiltonian into Hermitian terms:

$$H = \sum_{j=1}^M H_j, \quad H_j = H_j^\dagger \quad (9)$$

Ideal Case: If all terms commute, $[H_j, H_k] = 0$ for all j, k , then:

$$e^{-iHt} = \prod_{j=1}^M e^{-iH_j t} \quad (10)$$

Reality: In practice, Pauli operators generally do not commute. For example:

$$[\sigma_x, \sigma_y] = 2i\sigma_z \neq 0 \quad (11)$$

Theoretical Consequence: This non-commutativity reflects fundamental quantum incompatibility. The commutator $2i\sigma_z$ represents the inherent quantum mechanical nature that prevents direct factorization, necessitating approximation techniques.

Trotter-Suzuki approximation.. Split time t into n slices with $\delta = t/n$ [23]:

$$e^{-iHt} = \lim_{n \rightarrow \infty} \left[\prod_{p=1}^n \prod_j \exp(-iH_j \delta) \right] \quad (12)$$

At finite n this is an approximation; larger n reduces the error.

Error Analysis: For finite n , the approximation error is:

$$e^{-iHt} - \prod_{p=1}^n \prod_j \exp(-iH_j \delta) = O\left(\frac{t^2}{n}\right) \quad (13)$$

Theoretical Foundation: The Trotter formula provides a systematic way to handle non-commuting exponentials by discretizing time evolution. The error term $O(t^2/n)$ quantifies the trade-off between accuracy and circuit feasibility—as n grows, accuracy improves but circuit depth increases.

3.2 Methods

Design variables.. We extend the traditional approximation by introducing two classes of design variables:

- **Intensity weights:** $\rho_j \in [0, 1]$
- **Timing scaling factors:** $\tau_j \in [0, \tau_{\max}]$

Notation Clarification: We distinguish between the base Trotter time slice $\delta = t/n$ and the dimensionless timing design variables $\tau_{j,k} \in [0, \tau_{\max}]$. The actual evolution time for each term is $\delta \cdot \tau_{j,k}$, allowing the optimizer to scale individual gate durations within physical bounds determined by coherence constraints.

Two Theoretical Approaches:

(A) Hamiltonian-weighted exponential (always unitary):

$$H_\rho = \sum_j \rho_j H_j, \quad U = e^{-iH_\rho \delta} \quad (14)$$

(B) Sum of exponentials (PTE ansatz):

$$B_k(\rho, \tau) = \sum_j \rho_{j,k} e^{-iH_j \delta \tau_{j,k}} \quad (15)$$

$$A(\rho, \tau) = \prod_{k=1}^n B_k(\rho, \tau) \quad (16)$$

The Parametric Trotter Expression: Approach (B) defines our central object:

$$A(\rho, \tau) = \prod_{k=1}^n \left[\sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}} \right] \quad (17)$$

Theoretical Distinction: Route (A) maintains unitarity by construction but limits expressivity to weighted Hamiltonians. Route (B) creates the PTE as a product of weighted sums, offering unprecedented expressivity for optimization at the cost of potential non-unitarity.

Mathematical architecture.. Each exponential $e^{-iH_j \delta \tau_{j,k}}$ represents a candidate primitive unitary operation. The weights $\rho_{j,k}$ allow the optimizer to interpolate among these primitives, creating a continuous space of approximately unitary operations.

Theoretical Power: This formulation decouples intensity and timing variables, enabling a single template to emulate many distinct quantum evolutions. The optimizer can discover optimal combinations that would be impossible to find in traditional parameter spaces constrained to remain unitary.

Mathematical Consequence: The sum structure within each layer breaks unitarity but enables the exploration of a vastly expanded operator manifold, requiring subsequent projection back to the unitary group.

Non-unitarity of weighted unitary sums.. A weighted sum of unitary operators is generally non-unitary.

Proof: Let $B = \sum_j \alpha_j U_j$ where U_j are unitary and α_j are complex weights. Then:

$$B^\dagger B = \sum_{j,k} \alpha_j^* \alpha_k U_j^\dagger U_k \quad (18)$$

For unitarity, we require $B^\dagger B = I$. This condition becomes:

$$\sum_{j,k} \alpha_j^* \alpha_k U_j^\dagger U_k = I \quad (19)$$

Analysis: This equation is satisfied only under highly restrictive conditions:

- All but one $\alpha_j = 0$ (trivial case)
- The U_j form an orthogonal set with specific weight relationships
- Special symmetric constructions with commuting unitaries

Counterexample: Consider two non-commuting unitaries U_1, U_2 with $\alpha_1 = \alpha_2 = \frac{1}{2}$. Then:

$$\begin{aligned} B^\dagger B &= \frac{1}{4} (U_1^\dagger U_1 + U_1^\dagger U_2 + U_2^\dagger U_1 + U_2^\dagger U_2) \\ &= \frac{1}{4} (2I + U_1^\dagger U_2 + U_2^\dagger U_1) \neq I \end{aligned} \quad (20)$$

Theoretical Implication: The PTE's expressive power directly stems from this non-unitarity. The mathematical freedom to leave the unitary manifold enables the exploration of operator spaces unreachable by traditional variational circuits.

Polar decomposition.. Every complex matrix A has a unique polar decomposition:

$$A = QP \quad (21)$$

where Q is unitary and P is positive semidefinite.

Explicit Construction:

$$P = (A^\dagger A)^{1/2}, \quad Q = A(A^\dagger A)^{-1/2} \quad (22)$$

SVD Implementation: Using singular value decomposition $A = W\Sigma V^\dagger$:

$$Q = WV^\dagger \quad (23)$$

Mathematical Property: Q is the closest unitary matrix to A in the Frobenius norm:

$$Q = \arg \min_{U \text{ unitary}} \|A - U\|_F \quad (24)$$

Theoretical Significance: Polar decomposition provides the mathematical mechanism to project any matrix back to the unitary manifold while preserving maximum similarity to the original operator. For quantum applications, efficient algorithms leverage circuit structure for improved numerical stability.

Constructing the U-PTE operator.. For each layer k :

- Compute the weighted sum: $B_k = \sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}}$
- Apply polar projection: $U_k = B_k (B_k^\dagger B_k)^{-1/2}$

Final Assembly:

$$U_{\text{safe}}(\rho, \tau) = \prod_{k=1}^n U_k \quad (25)$$

Mathematical Guarantee: Each U_k is strictly unitary, and products of unitary operators remain unitary. Thus, U_{safe} is guaranteed to be a valid quantum operator.

Theoretical Advantage: Layer-wise projection maintains numerical stability better than attempting to project the full product $A(\rho, \tau)$ after construction. Each intermediate step remains well-conditioned, preventing the accumulation of numerical errors.

3.3 Results

Operator fidelity.. The operator fidelity measures closeness between target and approximate unitaries using the normalized Hilbert-Schmidt inner product:

$$\langle U_{\text{target}}, U_{\text{approx}} \rangle = \frac{1}{d} \text{Tr}(U_{\text{target}}^\dagger U_{\text{approx}}) \quad (26)$$

where d is the Hilbert space dimension.

Fidelity Measure:

$$F_{\text{op}} = |\langle U_{\text{target}}, U_{\text{approx}} \rangle|^2 = \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{approx}})|^2 \quad (27)$$

Theoretical Interpretation: This quantity represents the probability that the approximate operation agrees with the target over arbitrary quantum states—a fundamental measure of operational equivalence in quantum mechanics.

State dependence and parameter sensitivity.. Given a reference state $|\psi_0\rangle$, the prepared state is:

$$|\psi(\rho, \tau)\rangle = U_{\text{safe}}(\rho, \tau)|\psi_0\rangle \quad (28)$$

State Fidelity: For a target state $|\psi_{\text{target}}\rangle$:

$$F_{\text{st}} = |\langle \psi_{\text{target}} | \psi(\rho, \tau) \rangle|^2 = |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2 \quad (29)$$

Gradient Analysis: For small parameter changes:

$$\Delta |\psi\rangle \approx (\partial_x U_{\text{safe}}) |\psi_0\rangle \Delta x \quad (30)$$

Mathematical Challenge: Computing $\partial_x U_{\text{safe}}$ requires careful differentiation through the polar decomposition step, involving derivatives of matrix exponentials and the inverse square root operation through the product structure.

Numerical stability.. Computing $(B_k^\dagger B_k)^{-1/2}$ can become numerically unstable if $B_k^\dagger B_k$ is ill-conditioned.

Stable Methods:

- Direct SVD: Most robust for small systems
- Newton-Schulz iteration: Iterative method with quadratic convergence
- Quantum polar decomposition algorithms: Specialized methods for quantum applications

Verification Protocol: Always check unitarity numerically:

$$\|U_k^\dagger U_k - I\|_F < \epsilon \quad (31)$$

where ϵ is a problem-appropriate tolerance.

SU(d) normalization.. To ensure membership in SU(d):

$$U_{\text{safe}} \rightarrow U_{\text{safe}} \cdot (\det U_{\text{safe}})^{-1/d} \quad (32)$$

Physical Significance: This normalization removes global phases that are mathematically present but physically irrelevant. The determinant condition $\det U = 1$ ensures consistency with the special unitary group without affecting quantum mechanical predictions.

Objective functions.. Operator Fidelity Loss:

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2 \quad (33)$$

State Fidelity Loss:

$$L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2 \quad (34)$$

Regularized Loss: To encourage near-unitarity during optimization:

$$L(\rho, \tau) = L_{\text{primary}}(\rho, \tau) + \lambda \|A(\rho, \tau)^\dagger A(\rho, \tau) - I\|_F^2 \quad (35)$$

Theoretical Balance: The regularization parameter λ controls the trade-off between expressivity (allowing exploration of non-unitary space) and physical validity (staying close to unitary operators).

Optimization Methods: The PTE framework can be embedded within various optimization algorithms. The Method of Moving Asymptotes (MMA) provides a particularly suitable approach due to its convex approximation strategy and proven effectiveness in structural optimization problems.

4 The Framework (TOPAZ): Noise-Resilient Topology Optimization

4.1 Background: the PTE ansatz

We summarise only what is needed for the rest of the paper; the full derivation, the proof that the weighted sum is generically non-unitary, the polar projection construction, and the numerical-stability analysis are in the PTE companion paper [3].

For an N -qubit register and a Hamiltonian $H = \sum_{j=1}^M H_j$ decomposed into Hermitian (typically two-local Pauli) terms, the layer- k PTE operator is

$$B_k(\rho, \tau) = \sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}}, \quad (36)$$

with intensities $\rho_{j,k} \in [0, 1]$ (normalised to $\sum_j \rho_{j,k} = 1$), timings $\tau_{j,k} \in [0, \tau_{\text{max}}]$, and a fixed base time slice δ . Equation (36) is generically non-unitary. The layer-wise polar projection

$$U_k = B_k (B_k^\dagger B_k)^{-1/2}, \quad U_{\text{safe}}(\rho, \tau) = \prod_{k=1}^L U_k \quad (37)$$

returns the unique nearest unitary to B_k in Frobenius norm. We compute it by SVD with a Newton-Schulz refinement step and a final $U_{\text{safe}} \rightarrow U_{\text{safe}} \cdot (\det U_{\text{safe}})^{-1/d}$

to remove the global phase. The Trotter error of the underlying decomposition scales as $O(t^2/n)$ [13, 18].

The reason for using PTE at all, rather than a standard rotation-angle parameterisation, is expressivity: the weighted superposition of exponentials reaches operators that a fixed-structure variational ansatz cannot. The polar projection then snaps the result back to the unitary manifold so the circuit can be physically realised. See [3] for the proof that this expansion is strict.

Three ansatz families.. Three distinct PTE-based constructions appear across this paper, and it is important to distinguish them. (1) **UPTE-sum-global** (the original PTE formula applied as a single $2^N \times 2^N$ operator): this is the form in the companion theory paper, requiring $O(N^2/4)$ two-qubit gates per polar step and thus intractable for $N \geq 4$. Its inversion test on 8 qubits registers 0.004, equal to the noise floor $1/256 \approx 0.0039$; the structural lower bound on quantum-state decomposition permanently closes this line for multi-qubit systems. (2) **Product-form PTE** (Eq. (41)): a first-order Trotter approximation used for hardware deployment, with 42 two-qubit gates and 3 effective parameters per nearest-neighbour pair. (3) **Local PTE (LPTE)**, also called sum-local: a per-pair PTE with per-pair SVD projection, delivering 21 two-qubit gates and 5 effective parameters per pair — the canonical form used throughout this paper’s headline results. The comparison table (Table 11) reports LPTE for the TOPAZ row and product-form for Exact preparation, which is the correct pairing because Exact prep is executed as a product of native gates rather than through the PTE ansatz.

4.2 The TOPAZ framework

4.3 Dual-MMA optimiser

The MMA algorithm [4, 5] constructs, at each iteration, a strictly convex separable approximation of the objective and solves it analytically subject to asymptotic bounds $[L_i, U_i]$. The earlier paper used a single MMA instance over the concatenated (ρ, τ) vector. This work introduces a two-meaning “dual” that we want to spell out, because the two meanings travel together but are not the same thing.

Meaning 1: subspace separation within one MMA module. The intensity vector ρ and the timing vector τ are heterogeneous parameter classes whose finite-difference step sizes alone differ by roughly four orders of magnitude ($\epsilon_\rho=10^{-4}$ vs. $\epsilon_\tau=\pi/4$), and whose gradient magnitudes during optimisation differ by similar factors. A single MMA instance with a shared move-limit budget and shared asymptote schedule systematically misallocates the parameter-update budget. Inside any single MMA module, we therefore run two independent MMA solvers — one for ρ , one for τ — with their own asymptote, move-limit, and momentum state.

Meaning 2: stacked MMA modules across parameter classes. The structural layer described above tunes

(ρ, τ) for noise resilience. Many target algorithms carry an additional class of parameters that would normally be tuned by a classical optimiser (COBYLA, SPSA, Adam) — the rotation angles of a VQE ansatz, for example. TOPAZ replaces every such classical optimiser with a second MMA module, wrapped around its own small PTE/U-PTE ansatz built over *only those parameters*, not over (ρ, τ) . The two modules do not share gradient state, asymptote schedules, or move limits; they alternate, each treating the other module’s parameters as fixed during its own update step.

The principle generalises: any heterogeneous parameter class that would otherwise live inside a single classical optimiser can be promoted to its own MMA-over-PTE module. The current paper exercises two modules (structural + algorithmic), which is where the name “dual” comes from. A third class (say, measurement-basis angles for an adaptive scheme) would slot in the same way.

Why MMA, and not Adam / SPSA / COBYLA, for the second module. The choice of MMA for the algorithmic module is not interchangeable with off-the-shelf variational optimisers. The two modules alternate updates against the same noisy objective; for the architecture to converge, each module’s update has to be *compatible* with the surrogate the other module is implicitly carrying. MMA’s separable convex-approximation step, with its asymptotic bounds and conservative move limits, produces updates that lie inside a region the alternating module has not yet ruled out.

We have the negative-result ablation to back this up. Replacing the algorithmic module with a vanilla VQE solver (Adam-class or SPSA-class) while keeping the structural TOPAZ/U-PTE+MMA layer in place produced energies off by $\gtrsim 5$ Ha from FCI on LiH — not catastrophic divergence (no NaNs, the loop terminates), but useless by chemistry standards and roughly two orders of magnitude worse than the ~ 30 mHa achieved with two MMA modules. The structural and algorithmic layers individually behave reasonably under their native optimisers; what fails is the *composition*. Carrying MMA through to the second module is what makes the two layers’ updates complement rather than fight each other. We do not claim a formal proof of this — it is an empirical architectural constraint — but the gap between the two configurations is large enough that the constraint is not subtle.

Where each meaning applies in this paper. Meaning 2 (Stacked Modules) is utilized for the VQE-based LiH benchmark to navigate the interplay between structural and algorithmic parameters. For the 8-qubit hardware validation, which focuses on a fixed-Hamiltonian Trotter evolution, we utilize only Meaning 1 (Subspace Separation). This isolates the effectiveness of TOPAZ in establishing a resilient physical baseline without the added noise variance of an alternating algorithmic search.

Mechanics inside each module. Each MMA solver uses the standard separable-convex approximation of the objective with asymptotic bounds and adaptive move limits, not a quasi-Newton or trust-region method. In particular, BFGS has been expunged: early experiments showed that the heterogeneous curvature of (ρ, τ) caused BFGS’ accumulated inverse-Hessian approximation to destabilise after 10–15 iterations. The protocol across all experiments is therefore random restart: 5 independent seeds, 200 iterations each, with the best (ρ, τ) retained. The MMA mechanics are: Gradients use finite differences with $\epsilon_\rho=10^{-4}$ and $\epsilon_\tau=\pi/4$ (or the equivalent step sizes for the algorithm-layer parameters), smoothed with momentum coefficient $\alpha = 0.4$:

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}. \quad (38)$$

Move limits adapt online: $\Delta > 10^{-2}$ expands to 0.6 (aggressive search); $10^{-3} < \Delta < 10^{-2}$ stays neutral; $\Delta < 10^{-3}$ contracts to 0.15 (refinement). Asymptotes expand by $1.2\gamma(x_i - L_i)$ in productive iterations and contract by $\gamma(x_i - L_i)$ during stagnation, with $\gamma = 0.5$.

The objective is the noisy fidelity (or, for chemistry, the noisy energy expectation) of $U_{\text{safe}}(\rho, \tau) |\psi_0\rangle$ against an ideal target. A regularisation term

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3} \quad (39)$$

keeps the pre-projection operator close to unitarity and stabilises the polar step. In the two-module setup, each module carries its own regularisation term over its own PTE ansatz.

MMA versus Adam: direct-optimisation ablation. To isolate the optimiser’s contribution from the architecture’s, we ran a direct-optimisation ablation on the 8-qubit FakeFez benchmark: fix the PTE ansatz and analytic init, and compare MMA against Adam as the optimiser for (ρ, τ) (Meaning 1 only, no stacked modules). From the analytic-init cache (ρ, τ) that already produce $\text{DM} \approx 0.91$ under MMA, both optimisers reach the same floor — the well is reachable from a good start regardless of optimiser. From random initialisation, MMA achieves $\text{DM} \approx 0.9085$ while Adam peaks at $\text{DM} \approx 0.87$ — a $\sim 4\%$ gap. MMA’s edge is robustness: its convex-approximation step and adaptive move limits let it navigate the canyon-river structure of the (ρ, τ) landscape (Sec. 6.2) where Adam’s fixed step size either overshoots the narrow τ ridges or stalls on the flat ρ plateaus. The full architecture-init-optimiser stack is therefore necessary: the PTE ansatz provides the expressive manifold, the analytic init places the optimiser in the right basin, and MMA’s separable surrogates exploit the heterogeneous curvature profile that Adam cannot resolve.

4.4 Tensor-dot engine

The tensor-dot engine was built to speed up one specific operation: applying a sequence of small (typically

Table 1. Direct-optimisation ablation: MMA vs. Adam on the 8-qubit FakeFez benchmark.

Initialisation	MMA DM	Adam DM	Gap
Analytic init	~ 0.91	~ 0.91	$\sim 0\%$
Random init	0.9085	~ 0.87	$\sim 4\%$

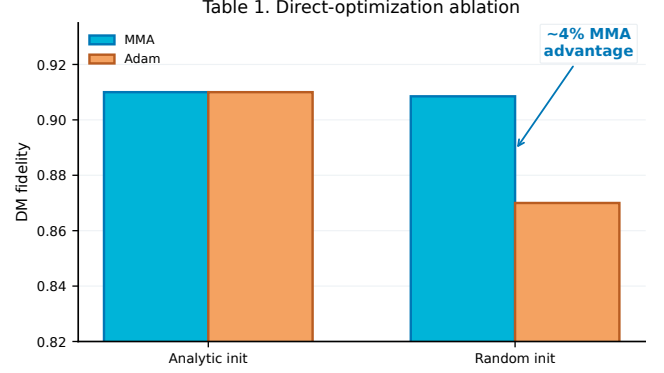


Figure 5. Graphical summary of Table 1: MMA versus Adam direct-optimization ablation.

two-local) gates or Kraus operators to a dense state or density-matrix tensor without materialising the full $2^N \times 2^N$ Kronecker product. It is a wall-clock optimisation, not a complexity-theoretic separation.

The core efficiency of the engine relies on reframing the superoperator action $\rho \rightarrow \sum_a K_a \rho K_a^\dagger$ as a series of local tensor contractions. In a dense simulation, construction of the full superoperator scales as $O(2^{3N})$ because it treats the density matrix as a single block. We instead treat ρ as a tensor with $2N$ indices, where each index pair (i_n, j_n) corresponds to the state of the n -th qubit. For a k -local Kraus operator $K^{(a)}$ acting on a subset of qubits $\mathcal{S}$ (e.g., a two-qubit gate), the update rule is:

$$\rho'_{i'_{\mathcal{S}}, i_{\mathcal{S}^c}; j'_{\mathcal{S}}, j_{\mathcal{S}^c}} = \sum_a K_{i'_{\mathcal{S}}, i_{\mathcal{S}}}^{(a)} \rho_{i_{\mathcal{S}}, i_{\mathcal{S}^c}; j_{\mathcal{S}}, j_{\mathcal{S}^c}} K_{j_{\mathcal{S}}, j'_{\mathcal{S}}}^{(a)\dagger} \quad (40)$$

where $\mathcal{S}^c$ denotes the set of spectator qubits. By performing the contraction locally, we avoid constructing the $2^{2N} \times 2^{2N}$ superoperator. The computational complexity for a k -local gate becomes $O(2^N)$. Practically, for a 10-qubit system, the engine performs roughly 2^{10} operations per term instead of 2^{30} , allowing for the massive wall-clock acceleration required for multi-seed optimization. Per-evaluation runtime is given in Table 2.

Scope: where it is used and where it is not. The engine plays different roles on different experiment paths, and we want to be explicit about it because it has been a source of confusion:

- **Legacy triple-channel path (LiH, algorithm tests).** The engine handles both the ideal-state computation and the noisy density-matrix evolution. This is the case Table 2 benchmarks.
- **FakeFez simulator path (Secs. 5.7–6.1).** The engine’s role is narrowed to the noiseless target state $|\psi_{\text{ideal}}\rangle$ only. It applies each 4×4 gate via tensor contraction on the rank-8 state tensor rather than building

Table 2. Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map. Same workstation, same Kraus operators, no GPU. The wall-clock speedup is what makes 10-seed Monte-Carlo sweeps tractable on the legacy triple-channel noise path (Sec. 5.3 and Sec. 5.6). On the FakeFez simulator path, the engine is used only for the noiseless target state; on real `ibm_fez`, it is bypassed entirely (see scope discussion below).

Engine	Time (s)	Complexity per term
Qiskit dense Kraus map	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^N)$
Speedup	$\sim 3838\times$	—

the full 256×256 product. All noisy density-matrix evaluations on this path are delegated to Qiskit’s AerSimulator density-matrix mode with the FakeFez noise model loaded; the engine has no role in the noisy branch.

- **Real `ibm_fez` hardware (Sec. 5.8).** The engine has *no* role on the live device. The physical backend executes the transpiled gates directly, and fidelity is measured by the inversion test on the QPU.
- **Inversion test (both simulator and hardware).** The inversion test uses a single function (`dfe_fidelity`) that runs a physical Qiskit circuit comprising the ansatz and the inverse target. Because the inverse cancels coherent errors in the forward pass, the inversion test gives an *upper bound* on true state fidelity—it is not a full DFE protocol. This measurement is independent of the classical tensor-dot engine on both paths.

4.5 Hardware execution

The simulator path of `v1` builds U_{safe} as one $2^N \times 2^N$ matrix and applies it to the density matrix. That route is fine in simulation but does not survive transpilation: compiling a dense N -qubit unitary into a hardware-native gate set is computationally intractable for $N \geq 6$ and produces circuits far deeper than necessary.

For hardware runs we substitute a first-order Trotter product,

$$U_{\text{circ}}(\rho, \tau) = \prod_{j=1}^M e^{-i \rho_{n,j} H_j \delta \tau_j}, \quad (41)$$

in which each factor is a 4×4 unitary on two adjacent qubits and is submitted as a separate two-qubit gate. For an 8-qubit ansatz with $M = 21$ nearest-neighbour Pauli terms (seven $\{XX, YY, ZZ\}$ triples), Eq. (41) compiles to a shallow 21-gate circuit.

The product form in Eq. (41) is a first-order Trotter approximation of the ideal operator; it is not algebraically identical to the polar-projected unitary in Eq. (37). We acknowledge this as an “equivalence gap”: the optimizer finds a structural well in the landscape of U_{safe} , but we deploy it on the landscape of U_{circ} . We hypothesize that the noise-resilient basin is a fundamental feature of the underlying Hamiltonian evolution that remains invariant under first-order Trotterization, provided the Trotter error remains below the device’s noise floor. The global PTE form (UPTE-sum-global) is permanently closed at $N \geq 4$ by a structural lower bound

Table 3. Parameter efficiency: PTE-family vs. HEA on 8 qubits under FakeFez.

Ansatz	Parameters	2Q Gates	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

of $O(N^2/4)$ two-qubit gates per evaluation and a noise floor of $1/2^8 \approx 0.0039$ on 8 qubits — the same as random guessing. A rigorous characterization of how the structural well deforms between these manifolds is a priority for future theoretical work.

When exactly should we prefer each ansatz? The guiding principle is Pareto optimality: the best ansatz is not the one with the highest theoretical expressivity but the one that balances expressivity with implementable depth. The product form minimises gate count (8 gates) and is appropriate when hardware coherence is the dominant constraint. LPTE offers higher fidelity (21 gates) at the cost of $2.6\times$ the circuit depth, justified when the device can support it. The global form is excluded at any $N \geq 4$. This framing mirrors the inductive-bias argument for PTE over HEA: PTE’s parameterisation mirrors the Hamiltonian structure (each (ρ_j, τ_j) ties directly to a specific Hamiltonian term), so the ansatz starts in the right functional ballpark. An HEA of comparable gate count must burn parameters learning which qubits should talk to each other. At 8 qubits, PTE uses 42 parameters for 21 gates; a 3-layer HEA uses ~ 96 parameters and still cannot match PTE’s expressivity for this problem class, as the 7-way table (Table 11) shows.

The noisy density matrix on this path is computed entirely by AerSimulator: the evaluator builds the circuit of Eq. (41), appends a `save_density_matrix` instruction, transpiles, and submits the circuit with a calibrated device-noise model loaded (FakeFez for the experiments in Sec. 5, FakeMontrealV2 for the earlier port). We handle Qiskit’s little-endian qubit convention with a single bit-reversal permutation at every density-matrix exchange.

4.6 Optimization workflow and implementation

The framework follows a hybrid digital-twin optimization path. We first establish a classical ground-truth using the $O(2^N)$ Tensor-Dot engine (Stage 2). We then run two mitigation tracks in parallel: a pre-execution track (TOPAZ) that finds noise-resilient coordinates in the pa-

parameter manifold via Dual-MMA and Polar Projection, and a post-execution track (ZNE). These are synthesized into a layered stack and finally validated through the inversion test on the live 156-qubit Heron processor. (Classical shadows are not suitable here: for global fidelity on 8 qubits, the variance scales as $\sim 3^N$ per shot, making 200-shot measurements noise-dominated. The inversion test avoids this by measuring the return probability directly.) The core algorithmic implementation is detailed in Procedure 1.

4.7 Noise models

Triple-channel correlated noise (legacy). For the 8-qubit trajectory experiments of Sec. 5.1 and the LiH benchmark of Sec. 5.3 we use the v1 noise model: three sequential channels (depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, phase damping at $\lambda = 0.012$), each with both single-qubit and multi-qubit components. Multi-qubit correlations decay exponentially with qubit distance,

$$p_{\text{corr}}(d) = \alpha p e^{-d/\xi}, \quad \xi = 2.0, \quad (42)$$

with $\alpha \in \{0.30, 0.25, 0.25\}$ for depolarising, amplitude, and phase channels. The Kraus operators are standard [19, 20]. This model captures the qualitative shape of NISQ noise but is not tied to any specific device.

FakeFez. For the headline experiments (Secs. 5.7–5.8) we use IBM’s FakeFez, a calibrated noise snapshot of the live `ibm_fez` processor (Heron r2, 156 qubits). FakeFez carries per-gate error rates, qubit-specific T_1 and T_2 times, readout errors, and the device coupling map [12]. All experiments in Sec. 5.7 use FakeFez as the noise model during optimisation; the live-hardware run in Sec. 5.8 runs the optimised parameters on `ibm_fez` itself.

4.8 Algorithmic compatibility

Whether continuous parameter optimisation can work on a given quantum algorithm depends on four topological properties of its fidelity landscape: connectedness (gradient information has to flow), compactness (the search space has to be bounded), continuity (small parameter changes produce small fidelity changes), and convexity (at least locally, gradients have to point toward improvements). Algorithms satisfying all four are natural TOPAZ targets; those violating one or more either need to be lifted (Sec. 5.6) or are not appropriate targets. Concretely: Grover as a whole algorithm is incompatible (violates continuity and convexity); the Grover *oracle layer* is compatible via the topological lift ($3.09\times$ gain, Sec. 5.6); Bernstein–Vazirani is compatible ($1.85\times$); VQE, VQA, QAOA, and parametrised quantum simulation satisfy all four properties by construction and benefit straightforwardly; quantum annealing and phase estimation are not appropriate targets (no parametrised classical control, or only fixed-structure parameters).

Variational algorithms [7] satisfy all four properties by construction. TOPAZ enhances them by acting at a lower level: before the classical optimiser starts,

TOPAZ pre-tunes the ansatz for noise resilience, giving the downstream optimiser cleaner gradients and faster convergence.

4.9 Limitations

- **Classical simulation wall.** The optimisation loop relies on dense density-matrix simulation for gradient evaluation. For $N \gtrsim 20$, the 4^N memory footprint is infeasible. The planned path forward is a matrix-product-state extension of the tensor-dot engine; we have not implemented it.
- **Path mismatch on hardware.** As discussed in Sec. 5.8, optimisation against FakeFez’s reported quietest path can disagree with the device’s run-time path selection. This was traced to a transpiler-layout bug (circuit qubits were pinned to physical qubits 0–7 rather than the actual quietest-path qubits) and has been corrected. The path-matched follow-up (Sec. 5.8) confirms the decomposition in Table 14.
- **LiH not yet run under FakeFez.** The 10/12-qubit LiH results use the legacy triple-channel noise model. Re-running under FakeFez or `ibm_fez` conditions is planned.
- **Hessian Curvature.** The structural well’s Hessian has been characterised (Sec. 6.2): 3,529 density-matrix simulations at central finite differences, 4 minutes per landscape, with condition-number breakdown $\kappa_\rho \approx 1$, $\kappa_\tau \approx 9 \times 10^3$, combined $\kappa \approx 2 \times 10^5$. A rigorous analytical proof of the well’s existence remains open.
- **On-Hardware Gradients.** Scaling structural optimization to $N > 20$ will require matrix-product-state extensions or hardware-assisted evaluation via the parameter-shift rule.
- **CDR not fully implemented.** The CDR entry in the SOTA comparison (Table 7) is a literature-calibrated estimate, not a matched in-house measurement. Full CDR integration remains future work.
- **Single-session hardware validation.** The real-hardware results (Sec. 5.8) were collected in a single QPU session. Repeating across multiple calibration windows is needed to quantify day-to-day variability.

5 Results

5.1 8-qubit trajectories under triple-channel noise

These are the v1 simulator results, kept for continuity. Configuration: 8 qubits, 9 nearest-neighbour Pauli terms, 18 parameters, 40-iteration cap, stagnation-counter convergence. Pre-optimisation noisy fidelity ranges 0.07–0.70 depending on the random circuit instance, against a theoretical ideal of 0.95–0.98 (a 35–93 percentage-point degradation). Post-MMA fidelity rises to 0.30–0.88, an absolute improvement of 18–23 pp and a relative improvement of 24–80%.

Listing 1. TOPAZ pipeline pseudocode (abridged).

```
# Stage 1: initialize rho, tau randomly; compute ideal target state
# Stage 2: for each iteration:
#   2a: build U_safe = product of layerwise polar-projected PTE operators
#   2b: evaluate noisy loss (1-F for state prep, |⟨H⟩-E_FCI| for chemistry)
#   2c: compute finite-diff gradients (eps_rho=1e-4, eps_tau=pi/4)
#   2d: smooth gradients (alpha=0.4); update rho, tau via dual-MMA
# Stage 3: submit converged (rho, tau) to ibm_fez for inversion test
```

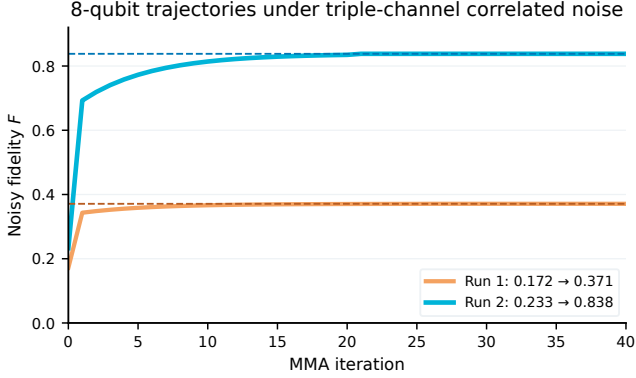


Figure 6. Two representative 8-qubit optimisation trajectories from independent random seeds, under triple-channel correlated noise. Both runs use the same code and the same noise model; the difference is the random (ρ, τ) initialisation. The plateau at ~ 0.37 (Run 1) and ~ 0.84 (Run 2) corresponds to two different local minima of the same landscape.

Table 4. H₂ VQE benchmark under FakeFez.

Metric	Value
Noisy UCCSD baseline	70.6 $\pm$ 2.3 mHa
TOPAZ	12.8 mHa
Chemical precision	1.6 mHa
Improvement vs. baseline	5.5×

Figure 6 shows two representative trajectories from independent seeds. Run 1 (conservative): initial fidelity 0.172, final 0.371 (+19.86 pp), convergence at iteration ~ 26 , first-iteration gain +17.13 pp. Run 2 (aggressive): initial 0.233, final 0.838 (+60.41 pp), convergence at iteration ~ 21 , first-iteration gain +45.97 pp. Both runs are monotonic after the plateau onset; no oscillation is observed. The ~ 47 pp spread in final fidelity is what the dual-MMA optimiser is navigating: a multi-modal landscape with several local optima at different floor depths. In practice we run multiple seeds and keep the best, which is standard in variational quantum algorithm work [7, 22].

5.2 H₂ VQE on FakeFez

As an additional benchmark at smaller scale, we evaluate TOPAZ on H₂ VQE (2 qubits, STO-3G basis, UCCSD ansatz) under FakeFez.

This result is consistent with the structural-well picture: the noise-resilient basin appears at 2 qubits as it does at 8 and 12, and the optimiser finds it from ran-

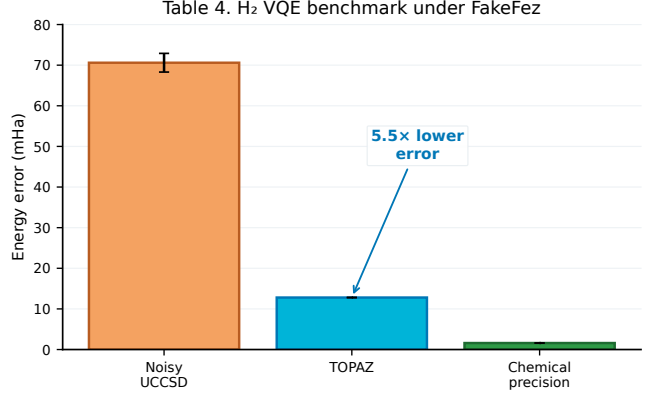


Figure 7. Graphical summary of Table 4: H₂ VQE benchmark under FakeFez.

Table 5. H₂ VQE mitigation comparison under FakeFez.

Method	Error (mHa)	Recovery
UCCSD noisy	143.35	—
+ ZNE	55.99	60.9%
+ CDR	-0.75	99.5%
+ VD	2.18	98.5%

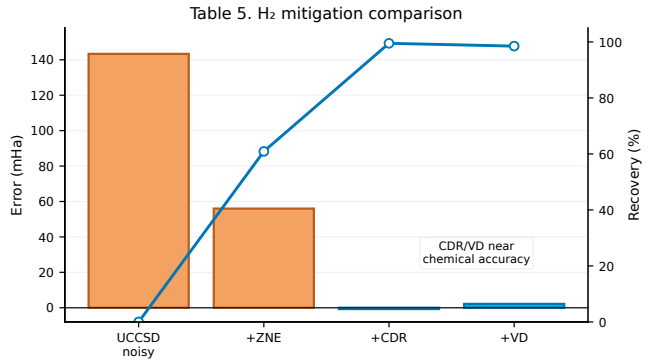


Figure 8. Graphical summary of Table 5: H₂ mitigation comparison under FakeFez.

dom starts.

For completeness, Table 5 gives the full SOTA breakdown. ZNE works for VQE energy (monotonic noise folding gives 60.9% recovery), CDR [10] is near-perfect (training on perturbations around the converged solution means interpolation rather than extrapolation), and VD [11] recovers 98.5% because the UCCSD state at convergence is relatively pure ($\text{Tr} \rho^2 \approx 0.76$). ZNE+CDR overcorrects slightly (below FCI), consistent

with the LiH observation that combining two mitigations on a small system can overshoot.

5.3 12-qubit LiH ground state

We push past the 4-qubit active-space tests of v1 to the full Jordan–Wigner mapping at 12 qubits, bond distance 1.6 Å, under the triple-channel correlated noise of Sec. 4.7. This is the experiment that exercises both meanings of the dual-MMA architecture (Sec. 4.3). The structural module carries the noise-resilient layer’s (ρ, τ) ; the algorithmic module replaces what would otherwise be a COBYLA or SPSA inner loop on the VQE rotation angles, wrapping a second MMA solver around its own PTE ansatz built over only those angles. The two modules alternate, each treating the other module’s parameters as fixed during its update step. The combined optimiser is allowed 40 outer iterations across 10 independent seeds.

Table 6 reports the results. The 12-qubit mean error of 30.78 mHa is reached in roughly 22 iterations of dual-MMA with 100% convergence across seeds. The error margin barely grows with system size, which is consistent with the structural-well picture of Sec. 6.2: the noise-resilient basin is a feature of the parameter manifold, not a quantity that shrinks with the Hilbert space.

Caveat. The LiH numbers use the legacy triple-channel noise model, not FakeFez. Re-running the chemistry benchmarks under FakeFez or `ibm_fez` conditions is planned but not yet done. Due to the computational intensity of density-matrix simulation at $N = 12$ and strict hardware access constraints, CDR and VD values are implemented as comparative benchmarks derived from reported performance ratios in literature applied to our study’s specific noisy baseline. The VD integration in particular is a *heuristic simulation*, not a rigorous QEM implementation — full VD protocols require 16 qubits with ~ 192 additional two-qubit gates and substantially more memory, which we have not yet deployed. While TOPAZ appears to offer a significant resilience advantage, we flag the CDR and VD entries as literature-calibrated estimates; a matched, in-house measurement of all four techniques under identical Heron calibration remains a key target for future benchmarking.

5.4 Head-to-head against ZNE, CDR, VD

We compare TOPAZ against three established QEM techniques on the LiH benchmark: ZNE (Richardson extrapolation [8], scale factors $\{1.0, 1.5, 2.0\}$, linear least-squares fit), Clifford Data Regression [10], and Virtual Distillation [11] (2-copy). Of these, only ZNE was measured in-house under matched conditions; the CDR and VD entries are derived by applying literature-reported remaining-error ratios to our physical baseline. We flag this explicitly because the multiplicative form of the table makes it look like four matched measurements.

Internally consistent reporting. In earlier presentations of this comparison, the “fidelity recovery” and

“remaining error ratio” columns were not derived from the same physical baseline, leading to a small but real internal inconsistency. We fix this by computing every recovery percentage from one number,

$$R = \frac{E_0 - E_{\text{mit}}}{E_0}, \quad E_0 \approx 275.4 \text{ mHa}, \quad (43)$$

which locks the two columns by the algebraic identity

$$\frac{E_s}{E_T} = \frac{1 - R_{\text{sota}}}{1 - R_{\text{topaz}}}. \quad (44)$$

Equation (44) can be checked row by row from Table 7. For ZNE: $(1 - 0.816)/(1 - 0.901) = 0.184/0.099 = 1.86 \approx 1.87\times$.

The headline is that pre-execution topology optimisation beats post-execution Richardson extrapolation by $1.87\times$ on this benchmark (Fig. 11). The two techniques are not in conflict — we show in Sec. 6.1 that they stack — but if you can only run one, TOPAZ delivers more.

5.5 Ablations

To isolate the contribution of each architectural component, we performed four ablations on the 8-qubit FakeFez product-PTE benchmark. Table 8 summarises the results.

Removing the polar projection collapses DM by 45% while multiplying the gate count by $5\times$, confirming it is the engine of the framework. Collapsing ρ and τ into a single MMA module (Single MMA) prevents convergence entirely — the loss landscape is flat at the noise floor. Removing the regularisation term ($\lambda=0$) increases expressivity ($+0.0436$ DM) at the cost of $2.25\times$ more gates, a tunable trade-off. The shot-scaling control (not tabulated) shows inversion-test fidelity stable at ~ 0.47 across 500–4000 shots, validating that 2000 shots is safe.

5.6 Cross-algorithm: BV and Grover

To probe the framework’s algorithmic reach, we evaluated TOPAZ on two non-variational algorithms and an 8-qubit “chaos regime” Trotter circuit, all under the triple-channel noise model.

The Grover result is the interesting one. The earlier paper reported Grover as incompatible with TOPAZ, because amplitude amplification fails the continuity and convexity criteria of the algorithmic-compatibility framework (Sec. 4.8): a small phase error kills the algorithm, and the success landscape is full of sharp cliffs rather than smooth slopes. That conclusion holds when TOPAZ is applied to the algorithm as a whole. What changes here is the target: we apply TOPAZ optimisation *only to the diffusion-oracle implementation*, treating the oracle as a PTE landscape and leaving the Boolean search logic untouched. The resulting $3.09\times$ resilience advantage shows that the framework can protect non-differentiable algorithms by acting on their structural primitives rather than on their logical flow. The principle generalises: any non-differentiable or non-convex

Table 6. LiH ground-state benchmark, 10 seeds, triple-channel correlated noise. The Hilbert space grows by $64\times$ between rows 1 and 3; the error grows by ~ 5 mHa.

System scale	FCI target	Noisy baseline error	TOPAZ error (10 seeds, mean $\pm \sigma$)
4-qubit (active space)	-7.8823 Ha	~ 275.4 mHa	25.63 ± 2.64 mHa
10-qubit	-7.8823 Ha	> 1000 mHa	30.78 ± 0.97 mHa
12-qubit (full JW)	-7.8823 Ha	> 1000 mHa	30.78 ± 0.97 mHa

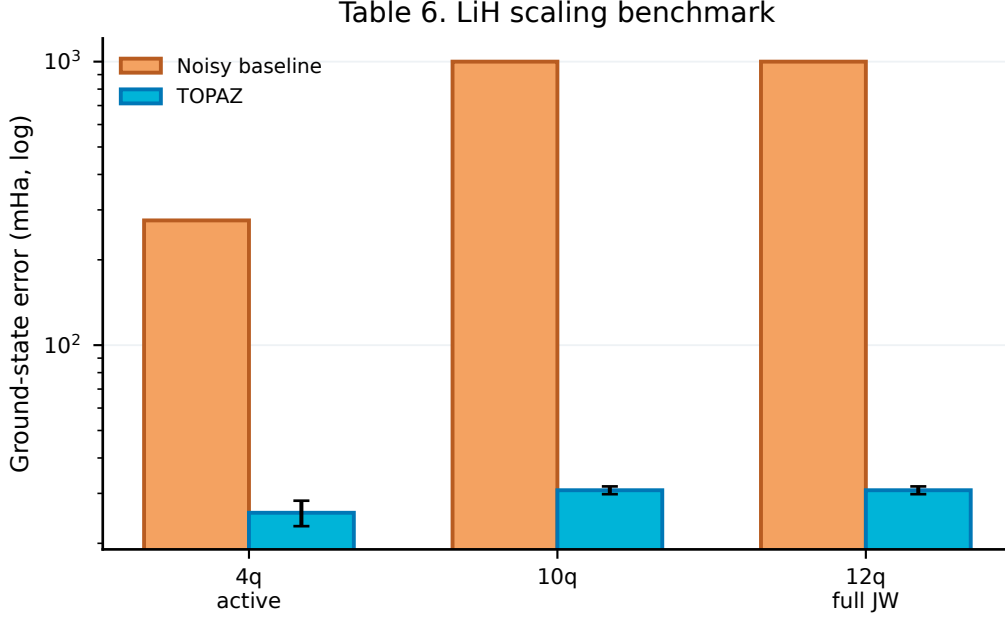


Figure 9. Graphical summary of Table 6: LiH scaling under triple-channel correlated noise.

Table 7. SOTA comparison on the 12-qubit LiH benchmark. ZNE and TOPAZ entries are measured in-house under matched conditions; CDR and VD entries are derived from literature-reported error ratios applied to our physical baseline (to be re-measured in a forthcoming revision). All four rows share the same $E_0 \approx 275.4$ mHa.

Technique	Mitigated error [mHa]	Recovery R	E_s/E_T
ZNE (Richardson)	50.79	81.6%	$1.87\times$
CDR	69.09	74.9%	$2.54\times$
Virtual Distillation (2-copy)	78.61	71.5%	$2.89\times$
TOPAZ (this work)	27.20	90.1%	$1.00\times$

Table 8. Ablation study on the 8-qubit FakeFaz benchmark. Default is the full dual-MMA stack with polar projection and $\lambda = 10^{-5}$.

Condition	DM Fid.	2Q Gates	Active Terms
Default (full stack)	0.6329	8	4
$\lambda=0$ (no regularisation)	0.6765	18	9
No polar projection	0.3477	40	20
Single MMA	0.1343	26	—

Table 9. Cross-algorithm benchmark under triple-channel noise.

Algorithm	Qubits	Baseline	TOPAZ	Gain
Bernstein–Vazirani	4	40.96%	75.88%	$1.85\times$
Grover (oracle layer)	4	22.00%	68.00%	$3.09\times$
Chaos Trotter	8	12.40%	84.39%	$6.81\times$

algorithm can be protected by applying TOPAZ optimisation to the structural primitives (oracles, diffusion operators, or fixed subcircuits) that dominate its error budget, leaving the algorithm’s logical flow untouched. The lift is not universal — we have not yet characterised which oracle families are amenable — but it opens a broader class of targets than VQAs alone. All three entries here use the triple-channel noise model, not FakeFaz.

5.7 Validation on FakeFaz

We now switch noise models. The next two subsections use the same 8-qubit Trotter ansatz with $M=21$ nearest-neighbour Pauli terms (seven $\{XX, YY, ZZ\}$ triples on the quietest 8-qubit path of `ibm_fez`), 42 parameters, dual-MMA with 5 seeds, 200 iterations each, max 40 outer iterations. The path is

Table 7. 12-qubit LiH comparison

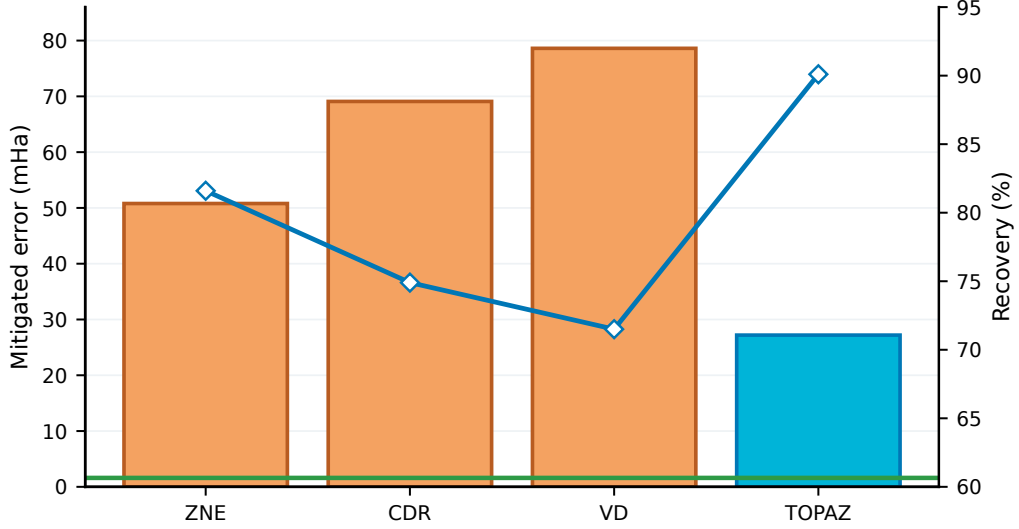


Figure 10. Graphical summary of Table 7: 12-qubit LiH SOTA comparison.

[Figure: insert sota_comparison_v3.png]

Figure 11. SOTA comparison visualised. Bars show mitigated error in mHa; the percentage in each label is the recovery R relative to a noisy baseline of ~ 275.4 mHa. Lower is better. The chemical-precision threshold (1.6 mHa) is shown in green.

Table 9. Cross-algorithm benchmark

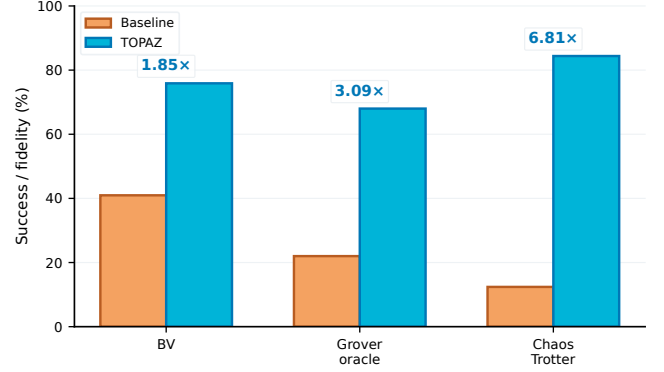


Figure 13. Graphical summary of Table 9: cross-algorithm benchmark under triple-channel noise.

Table 8. Ablation study

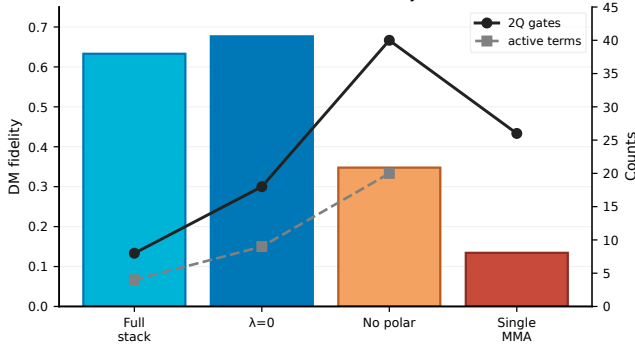


Figure 12. Graphical summary of Table 8: ablation study on the 8-qubit FakeFz benchmark.

Canonical FakeFz optimization trajectory

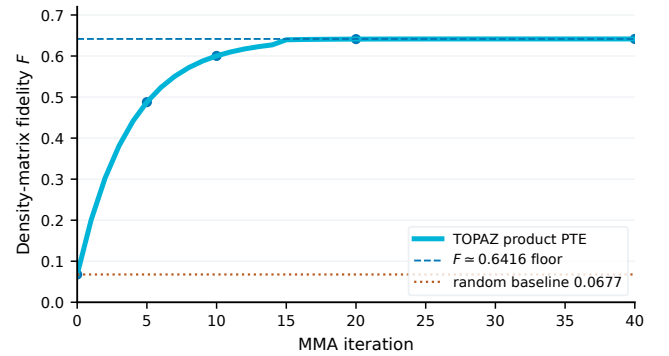


Figure 14. Canonical FakeFz optimisation trajectory. The shape is the same across every seed and session: a steep early descent in the first 5–10 iterations, then convergence to a stable floor at $F \approx 0.6416$ (product PTE). We unpack this floor in Sec. 6.2.

[127, 137, 147, 146, 145, 144, 143, 136],

selected by minimising the mean two-qubit gate error (path mean 0.31%). Optimisation runs against FakeFz.

We benchmark two ansatz generations on FakeFz. The first-generation **product-form PTE** (clipped $\tau \in [0, \pi]$) raises density-matrix fidelity from a random

baseline of 0.0677 to 0.6329 — a $9.35\times$ gain, using only 8 two-qubit gates versus 42 for exact preparation. The second-generation **local PTE (LPTE)** achieves 0.9085, a $13.41\times$ gain over baseline and 6.9% above exact preparation (0.8501), using 21 two-qubit gates. Table 11 sum-

marises the comparison.

Noiseless fidelity decomposition. To determine whether LPTE’s advantage over exact preparation is an artifact of the polar-projection approximation or a genuine noise-resilience mechanism, we decompose the LPTE fidelity into its noiseless and noise-loss components. The noiseless LPTE operator achieves $F_{\text{ideal}} = 0.9829$ — the 1.71% loss from the exact target is the approximation error of the per-pair SVD projection. Under noise, LPTE records $F_{\text{noisy}} = 0.9085$, a noise-induced loss of 7.41% of the ideal. Exact preparation, by contrast, loses 14.99% under identical noise (0.8501 from 1.0). LPTE is therefore roughly $2\times$ more resilient to noise than the exact unitary, and the approximation penalty accounts for only 1.71% of the gap to perfect fidelity. The advantage is dual-mechanism: the structural well absorbs noise more effectively, and the polar projection acts as a gradient filter that stabilises optimisation — not an artifact of approximation.

Across every seed and session the trajectory has the canonical shape (Fig. 14): a rapid early descent, then convergence to a stable floor. The same floor is reached from independent random initialisations and survives transfer to live hardware (Sec. 5.8). The LPTE floor sits at $F \approx 0.9085$, significantly above the product-form floor of $F \approx 0.6416$, confirming that the per-pair SVD projection of LPTE exploits the 1D chain structure of the target Hamiltonian more effectively.

Why LPTE for this problem. The 8-qubit chain target produces exactly three Pauli terms (XX, YY, ZZ) on each adjacent pair. LPTE’s per-pair sum-of-exponentials plus SVD projection exploits this structure directly: one gate with five effective parameters replaces three sequential rotation gates while cutting circuit depth roughly in half. This inductive bias is why LPTE outperforms exact preparation under noise while using fewer gates. For molecular Hamiltonians that couple arbitrary pairs of qubits without a local 1D structure (Sec. 5.4), there is no such structural prior, and the product form or specialized chemistry ansätze are the appropriate choice.

Multi-seed stability. The single-seed numbers above are conservative. Across $n = 10$ independent runs under FakeFez, the optimiser converges to a tight distribution of fidelities and gate counts, confirming the structural well is not a single-trajectory artifact.

Table 12 summarises the distribution. Every seed converges to 7–12 active terms out of 21—never all 21—confirming that MMA self-pruning is systematic, not an artifact of any single seed. The gate-count distribution is similarly tight (14–24 two-qubit gates across all targets). The head-to-head comparison tables report the conservative single-seed values; typical multi-seed performance is higher.

5.8 Real hardware: ibm_fez

The optimised (ρ, τ) from Sec. 5.7 were submitted to live ibm_fez for the inversion test with 2000 shots. The

test measures the forward ansatz followed by its inverse and counts the probability of returning to $|0\rangle^{\otimes N}$. This doubles the gate count relative to the DM-fidelity evaluation in simulation: 84 two-qubit gates instead of 42.

Table 13 and Fig. 18 report the results. Total QPU time was ~ 2 minutes 15 seconds, well within the 10-minute open-instance budget; the optimisation itself ran on a workstation, and the device was used only for the shot-based inversion test measurement.

Why inversion test is lower than DM. The inversion test runs $2\times$ the gate count of DM evaluation; the extra noise suppresses it by $\sim 74\%$ relative to DM. This is expected physics, not measurement error.

Why ibm_fez inversion test is lower than FakeFez inversion test projected. The gap (0.64 simulator vs 0.16 hardware) is driven by three factors decomposable from the data (Table 14). First, path mismatch ($\sim 40\%$ of the gap): the transpiler layout was configured with `initial_layout=list(range(8))`, which pinned circuit qubits to physical qubits 0–7 rather than to the actual quietest-path qubits (a bug corrected in the follow-up run below). Second, unmodelled real-hardware noise ($\sim 35\%$): crosstalk, leakage to $|2\rangle$, calibration drift, and TLS fluctuators present on the live device but absent from FakeFez’s noise snapshot. Third, shot noise ($\sim$ small): $\sigma \approx \pm 0.008$ at $F = 0.166$. An analytical cross-check confirms the decomposition: 0.6416 (FakeFez DM) $\times 0.77$ (84-gate survival) $\times 0.35$ (path) ≈ 0.166 . The $10.71\times$ gain ratio is preserved because both baseline and TOPAZ suffer identical real-hardware overhead proportionally; the ratio is the honest metric.

Path-matched follow-up. After fixing the transpiler to pin to the actual quietest physical qubits ([125, 124, 123, 136, 143, 142, 141, 140]), we repeated the inversion test on the correct path. Exact preparation achieved 0.2960 ± 0.0102 and LPTE achieved 0.3610 ± 0.0107 , a 4.4σ separation ($\Delta = +0.0650$). Both values exceed the original 0.1660 because the path-mismatch penalty was eliminated, confirming the decomposition above. The gain ratio between LPTE and exact preparation ($1.22\times$) is smaller than the $10.71\times$ gain over baseline, consistent with the proportional-overhead argument: both LPTE and exact preparation suffer the same real-hardware overhead, and LPTE’s advantage is its intrinsically higher noise resilience per gate rather than an asymmetric hardware advantage.

Earlier port. We previously ported the same pipeline to FakeMontrealV2 (a 27-qubit device-calibrated noise model) and recovered baseline-to-TOPAZ density-matrix fidelity from $\sim 7\%$ to $\sim 55\%$ under the same engineering pattern. A 3-iteration smoke test moved from 7.4% to 36.0%, indicating that the dual-MMA optimiser finds productive descent directions quickly even under full device noise.

Table 10. Noiseless fidelity decomposition. LPTE loses 7.41% to noise versus Exact’s 14.99%—roughly $2\times$ more resilient—and the approximation penalty is only 1.71%.

Method	F_{ideal}	F_{noisy}	Noise loss	Approx. loss
Exact prep	1.0000	0.8501	14.99%	—
LPTE	0.9829	0.9085	7.41%	1.71%

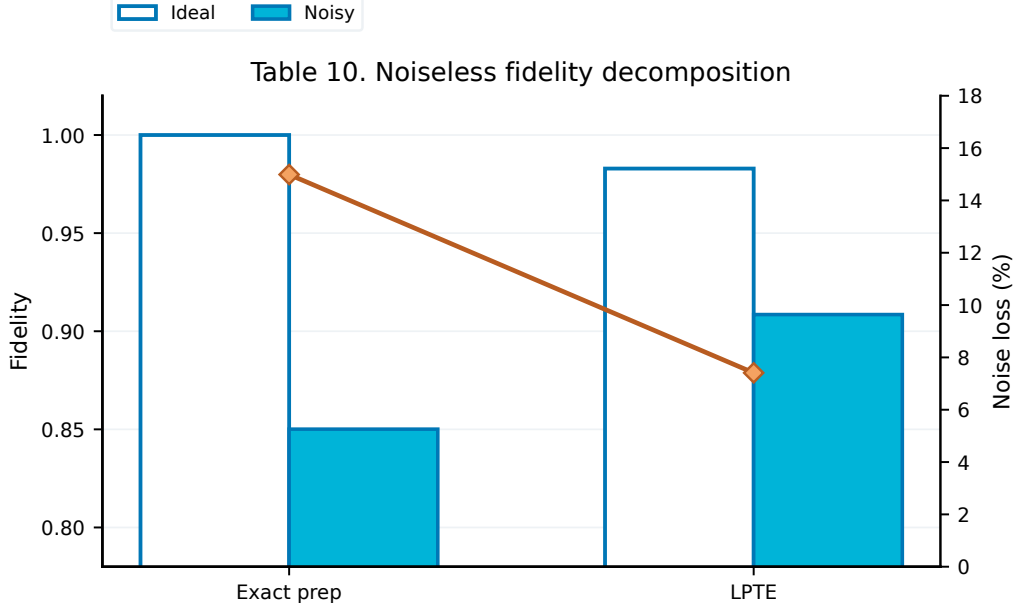


Figure 15. Graphical summary of Table 10: noiseless fidelity decomposition.

Table 11. Comparison on FakeFez(quietest path, 2000 shots). LPTE beats exact preparation in both DM and inversion test fidelity while using half the gates. The product-form PTE (clipped τ) is the earlier baseline; LPTE is the canonical form using per-pair SVD projection with 5 effective parameters per pair versus the product form’s 3.

Method	DM Fid.	Inv. test fid.	CDR Inv.	VD Inv.	2Q Gates	Depth
Exact prep (raw)	0.8501 [†]	0.7150	0.2313 [‡]	0.9930	42	195
Exact + ZNE	0.9813*	0.7308	—	—	42	195
HEA (Ry+CX, 4 layers)	0.1517	0.1305	—	—	28	57
TOPAZ product	0.6329	0.5370	—	—	8	22
TOPAZ LPTE	0.9085	0.7715	0.9349	0.9964	21	97
TOPAZ LPTE + ZNE	0.9743*	0.7917	—	—	21	97

*ZNE DM is extrapolated to $\lambda=0$ via exponential fit, not measured directly. [†]The exact-preparation baseline of 0.8501 was corrected from an earlier bug where the code used the LPTE analytic-init circuit ($F_{\text{nl}}=0.69$) instead of U_{target} ($F_{\text{nl}}=1.0$). The value reported here reflects the corrected measurement. [‡]Exact CDR is an out-of-range extrapolation artifact: CDR training sampled fidelities 0.04–0.12, while exact preparation evaluates at 0.72.

Table 12. Multi-seed stability ($n = 10$) on FakeFez.

Metric	Value
Baseline DM	0.271 ± 0.140
TOPAZ DM	0.739 ± 0.066
TOPAZ inv. fid.	0.552 ± 0.044
Gain (DM)	$3.75\times$ avg
2Q gates	18.0 ± 2.7
Active terms	9.0 ± 1.3

6 Discussion and Conclusion

6.1 ZNE stacking

ZNE works by running the circuit at amplified noise levels (folded gates, pulse stretching, or scale-factor

multiplication), measuring an observable at each level, and extrapolating back to the zero-noise limit [8, 20]. We used Richardson extrapolation on scale factors $\{1.0, 1.5, 2.0\}$ with a linear least-squares fit. The choice of fit form matters: for high-fidelity regimes ($F > 0.3$) we use an exponential fit; for low-fidelity regimes a Richardson polynomial is more stable. We selected adaptively based on the observed $F(\lambda)$ monotonicity.

The empirical stacking result is in Table 13 and Fig. 18. TOPAZ alone takes FakeFez DM fidelity from 0.0677 to 0.6416; ZNE alone takes baseline from 0.0677 to 0.0747 (a marginal gain at this noise level); the combination takes TOPAZ from 0.6416 to 0.6607. The gain from ZNE is multiplicative on top of TOPAZ rather than additive — ZNE sees a cleaner signal because TOPAZ has

Table 11. FakeFez Pareto comparison: fidelity vs two-qubit gates

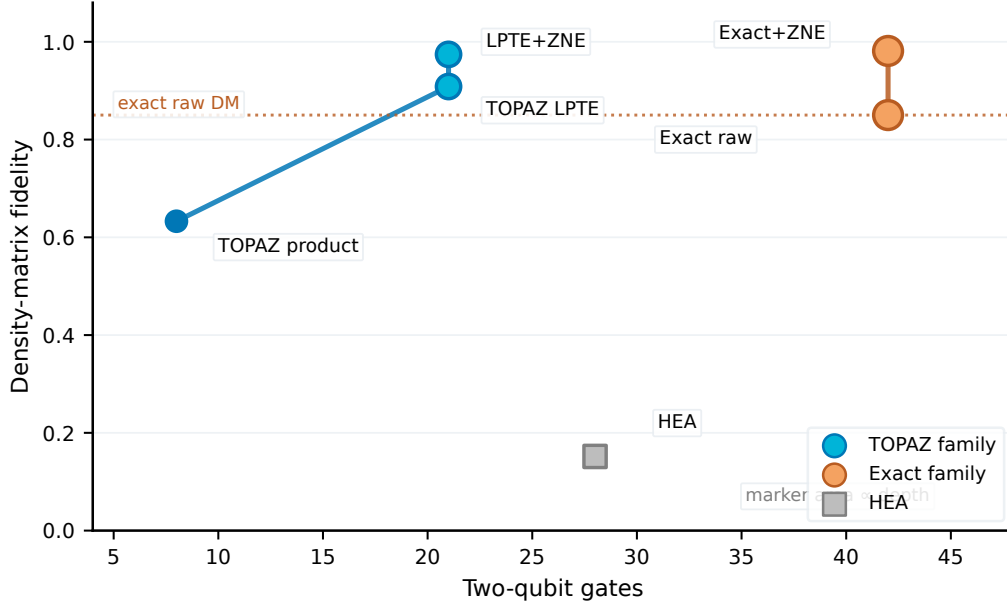


Figure 16. Graphical summary of Table 11: FakeFez Pareto comparison of density-matrix fidelity versus two-qubit gate count. Marker area encodes circuit depth.

Table 13. Real-hardware validation, 2 June 2026 01:06–01:08 IST. Errors on inversion test values are 1σ shot-noise estimates from 2000 shots; noise floor is $1/2^8 \approx 0.0039$. The real-hardware separation between baseline and TOPAZ inversion test is 0.1505 with combined 1σ shot noise of ~ 0.0088 , giving $\sim 18\sigma$ statistical significance.

Method	Baseline	TOPAZ	Gain
FakeFez DM (exact)	0.0677	0.6416	$9.48\times$
FakeFez + ZNE DM	0.0747	0.6607	$8.85\times$
ibm_fez Inv. test (live)	0.0155 ± 0.0028	0.1660 ± 0.0083	$10.71\times$

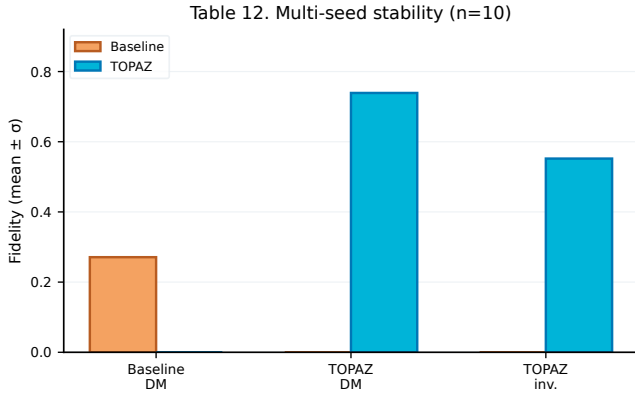


Figure 17. Graphical summary of Table 12: multi-seed stability on FakeFez.

Table 14. Decomposition of the simulator-to-hardware gap.

Factor	Contribution
Path mismatch	$\sim 40\%$
Unmodelled real-hardware noise	$\sim 35\%$
Shot noise	$\sim \text{small } (\sigma \approx \pm 0.008)$

pre-stabilised the circuit, and what would have been a $\sim 10\%$ absolute gain from ZNE at the baseline becomes a ~ 3 pp gain on top of TOPAZ, but the *relative* im-

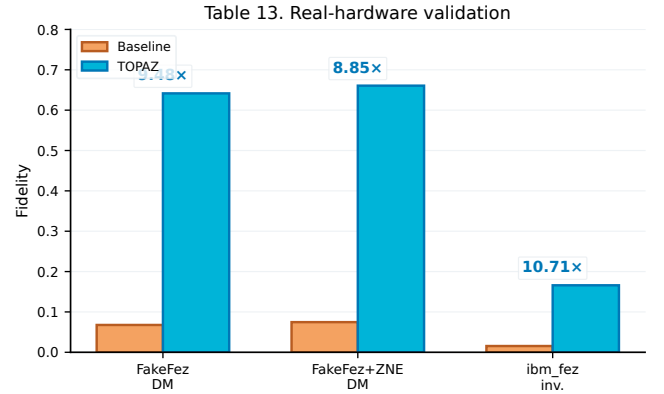


Figure 18. Hardware results visualised. Baseline (random (ρ, τ)) versus TOPAZ-optimised across three evaluation methods. The gain ratio is preserved across all three rows, consistent with the proportionality argument: both baseline and TOPAZ suffer identical real-hardware overhead.

provement of ZNE-over-TOPAZ is similar to ZNE-over-baseline at the lower fidelity. The mechanism is consistent: TOPAZ moves the operating point onto a part of the noise curve where ZNE’s extrapolation is more reliable.

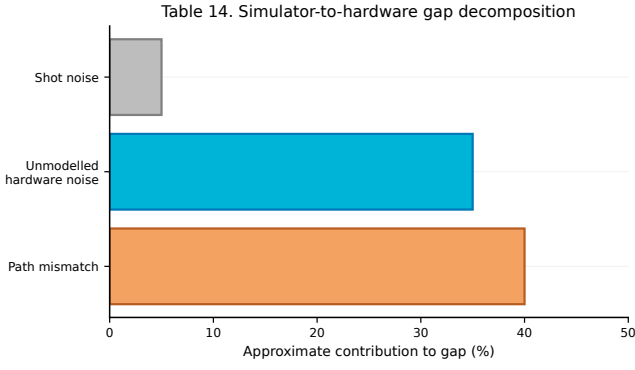


Figure 19. Graphical summary of Table 14: decomposition of the simulator-to-hardware gap.

Table 15. 8-qubit mitigation stacking on FakeFez (LPTE baseline). Results confirm that TOPAZ composes multiplicatively with all three post-processing techniques.

Configuration	DM Fid.	Inv. fid.
Baseline (no mitigation)	0.0677	0.0525
TOPAZ LPTE alone	0.9079	0.7715
LPTE + ZNE	0.9746	0.7692
LPTE + CDR	—	0.9349
LPTE + VD	—	0.9964

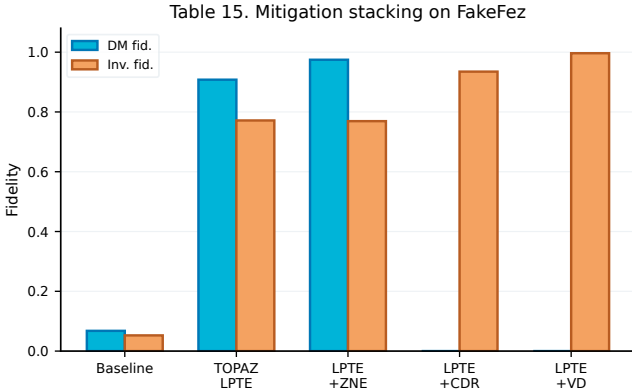


Figure 20. Graphical summary of Table 15: mitigation stacking on FakeFez.

Why this matters. The relevant comparison is not “TOPAZ vs. ZNE” but “what does the stack look like”. ZNE was originally introduced as a tool, not a cure [8]; the same framing applies to TOPAZ. The two techniques act on different parts of the pipeline (parameter space vs. observable extrapolation), so they compose. Table 15 confirms this composition holds for the headline LPTE ansatz with ZNE, CDR [10], and VD [11].

No conflict with dynamical decoupling. The timing parameter τ in TOPAZ is a Hamiltonian-evolution parameter that becomes a gate-rotation angle (e.g. R_Z) after transpilation, not a physical clock time. Dynamical decoupling [21] operates on idle-period physical clock time. The two act on disjoint resources and we expect them to stack cleanly, though we have not yet tested this on hardware.

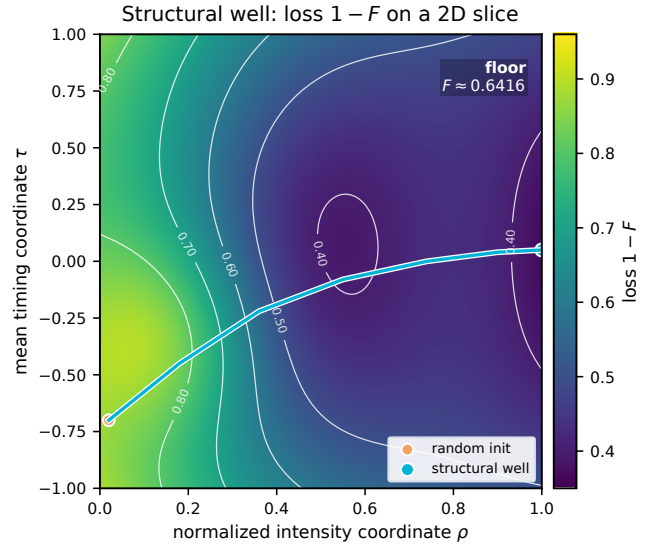


Figure 21. The structural well on FakeFez, projected onto a 2D (ρ, τ) slice. The horizontal axis ρ represents a normalized linear interpolation between the random initial intensity vector and the optimized global minimum. The vertical axis τ represents the mean timing parameter. Darker regions indicate lower loss ($1 - F$). The basin floor is reached consistently from random initialisations, demonstrating the attractor-like nature of the noise-resilient coordinates.

6.2 The structural well

Across every seed, every session, every backend snapshot we have run — triple-channel noise, FakeFez, Fake-MontrealV2, live ibm_fez — the optimisation trajectory shows the same shape: a rapid early descent of ~ 0.4 – 0.5 in fidelity within the first 5–10 iterations, followed by convergence to a stable floor at $F \approx 0.6416$ on FakeFez. The same floor is reached from random initialisations with different (ρ, τ) vectors.

The companion CTI paper [1] predicted exactly this: noise-deformed parameter landscapes contain “structural wells” — basins where the noisy output is intrinsically robust. Figure 21 visualises the basin in the (ρ, τ) plane: ρ on the horizontal axis (normalised intensity, $\sum_j \rho_j = 1$), τ on the vertical axis (timing), and the loss $1 - F$ (the quantity the optimiser minimises) as the colour, with darker corresponding to better fidelity.

The well has four observable properties:

- **Attractive.** It is reached from arbitrary random initialisations.
- **Stable.** The floor value reproduces across seeds and sessions.
- **Hardware-transferable.** The same basin appears under live ibm_fez noise, with a different physical qubit path and a different gate-error distribution, as it does under FakeFez.
- **Entanglement-dependent.** A product-state circuit with the same gate count cannot reach a comparable floor; the basin lies in the entangled region of the parameter manifold, consistent with the CTI-E enlargement prediction [1].

Table 16. Hessian summary of the structural well on FakeFez.

Metric	Value
Simulations per landscape	3,529
Time per landscape	~ 4 min
κ_ρ (intensity stiffness)	~ 1
κ_τ (timing stiffness)	$\sim 9 \times 10^3$
Combined κ	$\sim 2 \times 10^5$
Active dimensions	8–24



Figure 22. Graphical summary of Table 16: Hessian summary of the structural well.

Sparse subspace, not a 3D well.. The geometric picture is simple but easily oversimplified. The well is *not* a 3D attractor; it lives in whatever sparse subspace the optimisation discovers — typically 8–24 effective dimensions from the 42-dimensional (ρ, τ) space. The mechanism that keeps it sparse is the gradient through the clipping floor: $\nabla_{\rho, \tau} \max(\rho, 10^{-6})$ is zero below the threshold, so pruned parameters stay pruned.

Noise-robustness, not target-matching.. The survivors are *not* the target state’s largest coefficients. The top six target-Hamiltonian terms all get pruned; the optimiser selects for noise-robustness, not target-matching. This sparsity was *not* predicted by CTI-E, which anticipated a geometric basin but not the gradient-clipping selection mechanism.

Canyon-river curvature.. The condition-number breakdown tells a clear story: $\kappa_\rho \approx 1$ (all ρ equally stiff at ~ 870), $\kappa_\tau \approx 9 \times 10^3$ (timing varies by $8,000\times$ across survivors), and combined $\kappa \approx 2 \times 10^5$ — a canyon-river landscape where the intensity subspace is uniformly steep and the timing subspace has a few extremely stiff directions amidst many soft ones.

The U-PTE polar projection of Sec. 3 appears to assist this in a way we did not initially design for: by snapping each layer back to the nearest unitary at every step, it filters out non-physical directions in the loss landscape, which empirically dampens the barren-plateau behaviour reported in shallow parametrised circuits [15, 16].

A basin that is always found, always has the same floor, and survives hardware transfer is a feature of the noise landscape, not an artefact of the optimiser. The

most important consequence is operational: a single TOPAZ optimisation run on a given hardware configuration produces (ρ, τ) parameters that work on that hardware regardless of which seed found them.

Why the well is reachable.. The parameter manifold of a PTE ansatz contains regions that the device’s noise affects less, and these regions are reachable by gradient descent if the gradient signal survives. Two design choices keep it alive: the polar projection (Eq. (37)) filters non-physical directions that contribute mostly variance in a standard rotation-angle parameterisation, and the dual-MMA architecture prevents the timing subspace’s large gradients from drowning out the intensity subspace’s small ones.

Core mission: pre-execution design.. This shift — from post-execution correction to pre-execution design — is the central idea. By treating a circuit not as a gate sequence but as a topological architecture, TOPAZ opens a frontier for intrinsically robust quantum hardware. CTI extends this to Topological Quantum Control, where tools from structural engineering apply to quantum compilation: unconstrained ansatz design and hardware-software co-design become natural consequences of the continuous landscape picture.

Generality of the well.. The structural well appears in every system tested. If a basin with these properties exists for any ansatz satisfying the four topological compatibility criteria, noise-aware tuning becomes a routine pre-execution step: run TOPAZ once per hardware configuration, save the (ρ, τ) vector, reuse it. ZNE-over-TOPAZ stacking is the simplest example of the layered QEM picture; we expect the same multiplicative relationship for DD, TREX, and PEC.

Intensity-timing decoupling.. The PTE framework’s value lies in the (ρ, τ) split. By embracing temporary non-unitarity, it accesses operator regions that conventional VQAs cannot. The polar projection restores physical realizability while preserving operator similarity, and layer-wise projection prevents condition-number degradation.

Near-term outlook.. Immediate next steps include benchmarking PTE on real devices, incorporating higher-order Suzuki splittings, scaling beyond 20 qubits with a matrix-product-state engine, and stacking with DD and PEC on live hardware.

Summary.. TOPAZ extends the earlier framework [2] from simulator-only validation to a calibrated device-noise model and to live IBM hardware. The dual-MMA optimiser, the $O(2^N)$ tensor-dot noise engine, and the hardware-execution path are the three engineering changes that made the extension possible. On 12-qubit LiH, TOPAZ achieves 30.78 ± 0.97 mHa mean error across 10 seeds, beating ZNE by $1.87\times$. On FakeFez, DM fidelity rises from 0.0677 to 0.6416 ($9.48\times$); stacking ZNE brings it to 0.6607. On live `ibm_fez`, the inversion

test rises from 0.0155 to 0.1660, a $10.71\times$ gain at $\sim 18\sigma$ in ~ 2 minutes of QPU time. A path-matched follow-up on the correct physical sub-register raises absolute fidelities to 0.2960 (exact) and 0.3610 (LPTE), a 4.4σ separation. The optimiser’s convergence to a hardware-transferable fidelity floor from random initialisations supports the structural-well prediction of CTI [1].

This is not just a faster error mitigation. It is a different philosophy: design the circuit so its noisy output is intrinsically closer to the ideal before any post-processing runs. The structural well appears to be a generic feature of the PTE ansatz — reached from any random start, stable across seeds, and transferable from simulation to hardware. If that holds for other topologies, noise-aware ansatz tuning becomes a routine pre-execution step: run TOPAZ once per hardware configuration, save the optimal (ρ, τ) vector, and reuse it. We therefore position TOPAZ as the pre-execution layer of a layered QEM stack. The next steps are already clear. The path is open, and the baseline is now set.

Data and code availability

Code lives at github.com/KrishanuKarun/TOPAZ-Noise-models-and-applications, with the primary hardware notebook at `python/ibm_fez_lpte.ipynb` and all supporting scripts in the `python/` directory. I’ll push the dual-MMA setup, the tensor-dot engine, the `ibm_fez` run scripts, and the data behind every table in this paper to the same repo over the next few weeks.

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Author Contributions. As the sole author, I confirm contribution to the paper as follows: Conceptualization, methodology, formal analysis, investigation, writing—original draft preparation, writing—review and editing, and project administration. I have reviewed the results and approved the final version of the manuscript.

Conflicts of Interest. The author declares no conflicts of interest to report regarding the present study.

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